Non- Functional, System & Integration Requirements

1. **Non-Functional Requirements**

The following non-functional requirements were identified from the stakeholder engagement. The Big Brother Student Accommodation currently relies mainly on word-of-mouth advertising, student WhatsApp groups, taxi associations and group administrators to reach prospective tenants. Accommodation enquiries, bookings and payments are currently managed manually.

The proposed system should improve accessibility to accommodation information, simplify applications and bookings, protect student information, improve marketing and provide management with greater control over accommodation operations.

ID	Category	Non-Functional Requirement
NFR1	Security	The system shall require users to authenticate before accessing functions involving personal information, applications, bookings or account information.
NFR2	Security	Only authorized management users shall be allowed to access, monitor, analyze and manage student applications and accommodation information.
NFR3	Security	Student information and personal particulars shall be protected against unauthorized access.
NFR4	Security	Students shall only be able to view and modify information that pertains to their own accounts and applications.
NFR5	Security	Management shall only access student information for legitimate accommodation management purposes.
NFR6	Security	The system shall implement appropriate access controls

		to prevent unauthorized persons from viewing student information.
NFR7	Security	Development and testing shall use appropriate test data, and confidential stakeholder information shall not be uploaded to the public GitHub repository.
NFR8	Performance	The system should allow students to access accommodation information, check availability and submit applications without unnecessary delays under normal operating conditions.
NFR9	Performance	Accommodation availability, application and booking information should be retrieved efficiently.
NFR10	Usability	The application and website shall provide a clear and simple interface that allows students to easily understand the organization's vision, values and mission.
NFR11	Usability	Students shall be able to easily navigate to accommodation information, check availability, apply, make bookings and view status updates.
NFR12	Usability	The system shall provide clear instructions when students complete their particulars, applications and bookings.
NFR13	Usability	The system shall display clear error messages when information is missing, invalid or an application or booking cannot be completed.

NFR14	Accessibility	Accommodation information and application functionality should be easily accessible to prospective tenants without requiring them to rely solely on physical visits.
NFR15	Availability	Accommodation information and availability should be accessible whenever the application and website are available.
NFR16	Reliability	The system shall ensure that submitted student information, applications and bookings are stored correctly and can be retrieved by authorized users.
NFR17	Reliability	The system should reduce reliance on manual accommodation records by electronically storing relevant applications and booking information.
NFR18	Consistency	Accommodation, availability, application and booking information displayed through the Flutter application and ASP.NET website shall be based on the same backend information.
NFR19	Privacy	Student information shall be treated as confidential and shall not be accessible to unauthorized persons.
NFR20	Maintainability	Authorized management should be able to update accommodation information, availability and relevant business information without requiring major changes to the entire system.
NFR21	Marketing	The system should provide an effective digital platform

		through which the accommodation business can present its organization, accommodation, facilities and relevant information to prospective tenants.
NFR22	Payment	The system should provide an accessible payment process for accommodation where payment functionality is implemented within the approved project scope.

2. **System Requirements**
   The proposed Accommodation Application and Management System will consist of three main components:
   - **Flutter mobile application** – primary supporting students and prospective tenants.
   - **ASP.NET** web application supporting accommodation management.
   - **Supabase backend –** providing shared data storage and backend services.

SR1 – Flutter Mobile Application

The system shall provide a mobile application developed using Flutter for students and prospective tenants.

The Flutter application shall:

- Allow students and prospective tenants to register and log into the system.
- Allow students to view the organization's vision, values and mission.
- Allow students to view accommodation information.
- Allow students to view information about accommodation facilities.
- Allow students to view available accommodation.
- Allow students to check accommodation availability.
- Allow students to submit their personal particulars.
- Allow students view information pertaining to themselves.
- Allow students to add or update their own information where permitted.
- Allow students to apply for accommodation.
- Allow students to make accommodation bookings.
- Allow students to view their application and booking information.
- Allow students to view application and booking status updates.
- Allow students to receive relevant responses from management.
- Allow students to access relevant payment functionality where included in the approved project scope.
- Provide clear instructions during the application and booking process.
- Display confirmation messages when information is successfully submitted.

- Display appropriate error messages when an operation cannot be completed.
- Retrieve accommodation, availability, application and booking information from Supabase.

The Flutter application will give students easier access to accommodation information and allow them to complete important accommodation processes without depending entirely on physical visits, word-of-mouth advertising or informal communication channels.

SR2 – ASP.NET Web Application

The system shall provide an ASP.NET web application for authorized accommodation management.

The ASP.NET website shall allow authorized management to:

- Log into the management system securely.
- View and manage relevant organizational information.
- Manage accommodation information.
- Manage accommodation facility information.
- Manage accommodation availability.
- View prospective tenant information submitted through the system.
- View student particulars submitted during the application process.
- View accommodation applications.
- View accommodation bookings.
- Monitor application and booking statuses.
- Analyze information provided by prospective tenants.
- Respond appropriately to students and prospective tenants.
- Approve or reject applications/bookings where applicable.
- Update application and booking information where authorized.
- Manage and control student information according to the user's role.
- Monitor accommodation-related information and operations.
- Manage relevant payment information where payment functionality is included.
- Use student information to support appropriate accommodation decisions.
- Management information that is used to effectively market the accommodation business.
- Retrieve and update information through the shared Supabase.

SR3 – Supabase Backend

The system shall use Supabase as the shared backend for the Flutter mobile application and ASP.NET web application.

The Supabase shall be used to:

- ❖ Store student and prospective tenant account information.
- ❖ Store student particulars submitted during applications.
- ❖ Store accommodation information.
- ❖ Store accommodation facility information.
- ❖ Store accommodation availability information.
- ❖ Store accommodation application records.
- ❖ Store booking records.
- ❖ Store application and booking status information.
- ❖ Support user authentication.
- ❖ Support appropriate access controls.
- ❖ Store relevant payment information where payment functionality is implemented.
- ❖ Provide a shared data source for the Flutter application and ASP.NET website.
- ❖ Allow authorized management to access and manage relevant information.
- ❖ Maintain consistency between information accessed through the mobile application and website.

3. Integration Requirements

The three system components shall operate as one integrated accommodation management solution.

The overall system structure will be:

Flutter Mobile Application ⇔ Supabase Backend ⇔ ASP.NET Web Application

IR1 – Flutter to Supabase Integration

The Flutter mobile application shall communicate with Supabase to retrieve accommodation information and availability and to submit student information, applications and bookings.

The expected data flow is:

Student ➔ Flutter Application ➔ Supabase➔ Flutter Application ➔ Student

For example:

1. A student opens the Flutter application.
2. The student views the organization's vision, values and mission.
3. The student views accommodation and facility information.
4. The application retrieves accommodation information from Supabase.
5. The student checks accommodation availability.
6. The student registers or logs into the system.
7. The student enters or updates their personal particulars.
8. The student submits an accommodation application.
9. The application information is stored securely in Supabase.

10. The student makes a booking where appropriate.
11. Relevant booking information is stored in Supabase.
12. The student can view their application and booking status through the application.
13. The student can access relevant payment functionality where implemented.

IR2 – ASP.NET to Supabase Integration

The ASP.NET web application shall communicate with Supabase to allow authorized management to monitor, analyze, manage and respond to information submitted by students.

The expected data flow is:

Management ➔ ASP.NET Website ➔ Supabase ➔ ASP.NET Website ➔ Management

For example:

1. An authorized management user logs into the website.
2. The website retrieves student applications from Supabase.
3. Management reviews the information submitted by the prospective tenant.
4. Management analyses the information.
5. Management approves, rejects, updates or responds to the application where appropriate.
6. The relevant information is updated in Supabase.
7. The student can view the relevant status or response through the Flutter application.

This provides management with a centralized method of controlling, monitoring, analyzing and responding to student information.

IR3 – Shared Data Integration

The Flutter application and ASP.NET website shall use the same Supabase backend.

This integration shall ensure that:

- ❖ Students can view current accommodation information.
- ❖ Students can check current accommodation availability.
- ❖ Applications submitted through Flutter are available to authorized management through ASP.NET.
- ❖ Bookings submitted by students are available to authorized management.
- ❖ Students can view information relating to their own applications and bookings.
- ❖ Management can monitor student applications and bookings.
- ❖ Management can analyze and respond to information submitted by students.
- ❖ Application and booking status updates can be reflected in the student's application.
- ❖ Students can update their own information where permitted.

- ❖ Management can manage and control information according to their authorized role.
- ❖ Student information remains protected from unauthorized access.
- ❖ Relevant payment information can be processed through the integrated system if payment functionality is implemented.
- ❖ Information about the organization, accommodation and facilities can be maintained centrally.
- ❖ Both platforms use consistent accommodation and application data.

4. Requirements Prioritization

The requirements have been prioritized according to Must Have, Should Have and Could Have. The prioritization is based on the stakeholder's identified needs and the problems experienced by the accommodation business:

Must Have Requirements

ID	Requirement	Description	Reason
M1	User Registration and Login	Students and management must have appropriate accounts.	Required for controlled system access.
M2	Organization Information	Students must be able to view the organization's vision, values and mission.	Specifically identified by the stakeholder.
M3	Accommodation Information	Students must be able to view accommodation and facility information.	Addresses the current difficulty accessing information.
M4	Accommodation Availability	Students must be able to check accommodation availability	Specifically requested by the stakeholder.
M5	Student Information	Students must be able to submit their particulars.	Specifically requested by the stakeholder.
M6	Student Information Management	Student must be able to view and update information pertaining to themselves where permitted.	Directly addresses stakeholder requirement 16

M7	Online Application	Students must be able to apply for accommodation.	Core accommodation process.
M8	Accommodation Booking	Students must be able to make accommodation bookings.	Specifically identified by the stakeholder.
M9	Application/Booking Status	Students must be able to check application and booking status updates.	Specifically requested by the stakeholder.
M10	Management Access	Management must be able to access information provided by prospective tenants.	Direct stakeholder requirement.
M11	Management Control	Management must be able to manage and control student information.	Direct stakeholder requirement.
M12	Management Analysis	Management must be able to analyze information provided by prospective tenants.	Direct stakeholder requirement.
M13	Management Response	Management must be able to respond appropriately to student applications and information.	Direct stakeholder requirement.
M14	Information Security	Student information must be protected from unauthorized access.	Specifically identified by the stakeholder,
M15	Shared Supabase Backend	Flutter and ASP.NET must use the same Supabase backend.	Required for an integrated solution.
M16	Role-Based Access	Students and management must have appropriate levels of access.	Protects information and supports different responsibilities.

Should Have Requirements

ID	Requirement	Description	Reason

S1	Accommodation Search	Students should be able to search for accommodation.	Makes finding accommodation easier.
S2	Accommodation Images	Listings should include accommodation photographs.	Helps students understand the accommodation before viewing it physically.
S3	Online Enquiries	Students should be able to submit enquiries through the system.	Provides an alternative to physical visits and informal communication.
S4	Management Dashboard	Management should have dashboard showing applications, bookings and availability.	Improves monitoring and management.
S5	Digital Marketing	The website/application should support effective presentation and marketing of the accommodation business.	Addresses the stakeholder's difficulty reaching prospective tenants.
S6	Status Notifications	Students should receive notifications when their application or booking status changes.	Improves communication.
S7	Payment System	The system should provide an accessible accommodation payment process if included in the approved project scope.	The stakeholder identified the need for easier access to payment functionality.
S8	Business Information Sharing	Accommodation information should be easy to access and share with prospective tenants.	Directly addresses the stakeholder's marketing and information-sharing challenge.

Could Have Requirements

ID	Requirement	Description	Reason
C1	Advanced Online Payment	Students could complete additional payment functions online.	Could improve convenience but requires additional implementation considerations.
C2	Reviews and Ratings	Students could review and rate the accommodation.	Useful for future marketing and engagement.
C3	Advanced Reports	Management could generate reports on applications, bookings and occupancy.	Useful for future business analysis.
C4	Virtual Accommodation Tour	Students could view the virtual tours of accommodation facilities.	Could further reduce dependence on physical viewing.
C5	Favorites	Students could save preferred accommodation listings.	Convenience feature that is not essential to the core system.

5. Stakeholder Evaluation and Future Review

The stakeholder indicated that they would only be in position to properly review the system after it has been developed and implemented.

This means that the stakeholder's final evaluation of whether the system meets the organization's needs will take place after the prototype/system has been developed and made available for use.

The project team should therefore:

- Present the developed system to the stakeholder after implementation.
- Demonstrate the main student and management functions.
- Allow the stakeholder to test the relevant processes.
- Obtain stakeholder feedback on the system.
- Identify any functionality that requires improvement.
- Record stakeholder feedback and any requested changes.
- Use the feedback to determine whether further requirements or scope changes are necessary.

The stakeholder's post-implementation review can be used as evidence for validating whether the proposed system addresses the original problems identified during stakeholder engagement.

6. Requirements Summary

The stakeholder consultation identified that the accommodation currently relies heavily on word-of-mouth advertising, student WhatsApp groups, taxi associations and group administrators to reach prospective tenants. Students are encouraged to physically visit the accommodation to view the facilities before deciding whether to apply.

The stakeholder identified difficulties in:

- Reaching the intended number of students.
- Making accommodation information accessible.
- Sourcing prospective tenants.
- Easily sharing business information.
- Managing bookings and payments through a manual process.
- Protecting students from accommodation scams.
- Managing information provided by prospective tenants.

The stakeholder further specified that students should be able to:

- Understand the organization's vision, values and mission.
- View accommodation and facilities.
- Check accommodation availability.
- Submit their particulars.
- View and update information pertaining to themselves where permitted.
- Apply for accommodation.
- Make accommodation bookings.
- Check application and booking status updates.
- Access relevant payment functionality.

Management should be able to:

- Access information provided by prospective tenants.
- Analyze student information.
- Monitor applications and bookings.
- Respond appropriately to students.
- Manage and control student information.
- Manage accommodation and availability information.
- Use the system to support effective marketing and information sharing.

The **Must Have requirements** therefore focus on the core accommodation process, from providing information and checking availability through to applications, bookings and management of student information.

The **Should Have requirements** improve accessibility, communication, marketing and administration, while **Could Have requirements** provide opportunities for future development if sufficient time and resources are available.

The requirements will be reviewed with the stakeholder after the system has been developed and implemented. This will provide an opportunity to determine whether the system adequately addresses the stakeholder's identified needs and whether further improvements are required.